

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element I Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to structure your testing, data collection, and analysis but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own testing collection and reports, be sure to delete the instructions and delete the examples given in the template table.

You must test *all* high priority requirements to score a 5. After testing, you will connect the data you collected to the "success" or "failures" of your design. You will consider next steps for your design based on the data you just analyzed. Each test that was listed in your table for Element H (testing plan) was on its own row. Generate a report for each test, the results, and its analysis that includes the following:

- a) number and name of the test (as referenced in Element H)
- b) indication of whether the test is high, medium, or low priority based on your requirements (as reference in Element C)
- c) for physical tests, include an explanation of your process and the full set of raw data such that the following best practices for scientific testing and mathematical modeling will be evident:
 - creating experimental controls
 - defining independent and dependent variables
 - only changing one variable at a time and changing it in increments to find a trend
 - collecting at least FIVE trials for each variable increment
 - plotting data appropriately (e.g. scatter plot with trendline for continuous variables, bar charts/histogram for non-continuous variables, acknowledging and recording error as appropriate, and labeling random and systematic errors).

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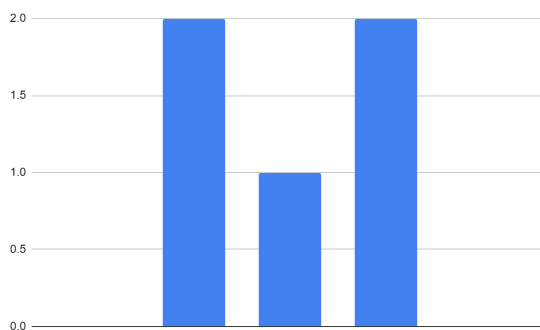
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- d) summary of your data with appropriate charts, graphics, spreadsheets, etc., drawing on best practices in science and math for data display and analysis
- e) statement of what the data means and a summary of the implications of the data you collected
 - o A possible sentence could be, "Based on our testing data, we found that <trend in data> means <something about the design's success> because <logic of conclusion>."
- f) discussions about *each* implication from the summary of your data
- g) next steps for your design and process from the new information from the test that was conducted

Tests:

1. How quick it opens: able to fully open and close in 5 sec: high priority
2. How far it opens: able to open in fully: High priority
3. Easily usable by students: give it to 3 students: High Priority
4. Visually appealing: Ask 3 students: Low Priority
5. Easily Understandable: Ask 3 students: Medium Priority

Test #1: tested 5 times: 4 seconds, 3 seconds, 2 Seconds, 4 Second, 2 Seconds



The Graph above represents the frequency of the number of seconds it takes to open and close the prototype. Out of 5 trials, 2 of them ended up being 2 and 4 seconds, and the other ended up being 3 seconds.

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- Based on our testing data, we found that the average open and close time is 3 seconds which means that the test was successful because it was under 5 seconds.
- The next step would be to reduce friction from the wire to make it even easier to open and close the machine.